PathFinder

An Indoor Navigation Web Application for the Physical Sciences Building in the University of the Philippines Los Baños

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Overview

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Problem Statement

New students and visitors of the campus often struggle to navigate buildings with multiple floors and classrooms. A physical floor plan exists, and there are also some campus maps available to view, but it's difficult to find and to navigate because it is in a fixed position.

Other existing solutions such as Global Positioning Systems (GPS) work well outdoors but struggle indoor navigation, which Indoor Positioning Systems (IPS) can fill this gap but require specialized hardware to set up and are expensive to maintain

Solution: Have a low-cost and scalable way to support digital indoor navigation without relying on external hardware per floor.

Objectives

01.
Integrate point to point directions to aid users in finding the exact locations of classrooms, offices, and facilities within the UPLB Physical Sciences Building.

02.
Enable building administrators to generate a building's maps based on floor plans, allowing for updates and customization of the mapping content.

03.
Assess the usability of the system through user testing to evaluate navigation accuracy, interface intuitiveness, and overall user satisfaction.

Main Objective:

Develop a web application that provides an indoor map of the UPLB Physical Sciences Building

METHODOLOGY

Requirements Gathering

Actual Layout of the Building

An email request was made to the University Planning and Maintenance Office (UPMO) to get the building's floor plans.

Admin Testers

Another email was sent to UPMO to request user testing. Administrators from their office were asked to participate in testing the layout builder feature of the application.

METHODOLOGY

User Testing

Regular Users

Tested navigation accuracy by using the indoor navigation user interface to select different origin and destination points

Building Administrators

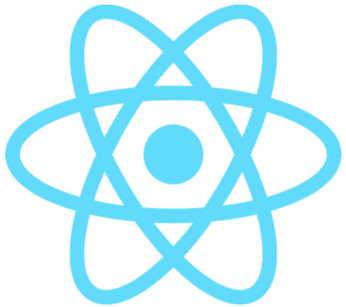
Tested the layout builder functionalities: add and edit building components, save as a draft, and publish to the main navigation page

METHODOLOGY

Technology Stack



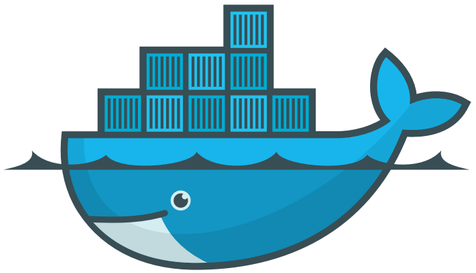
Python



React



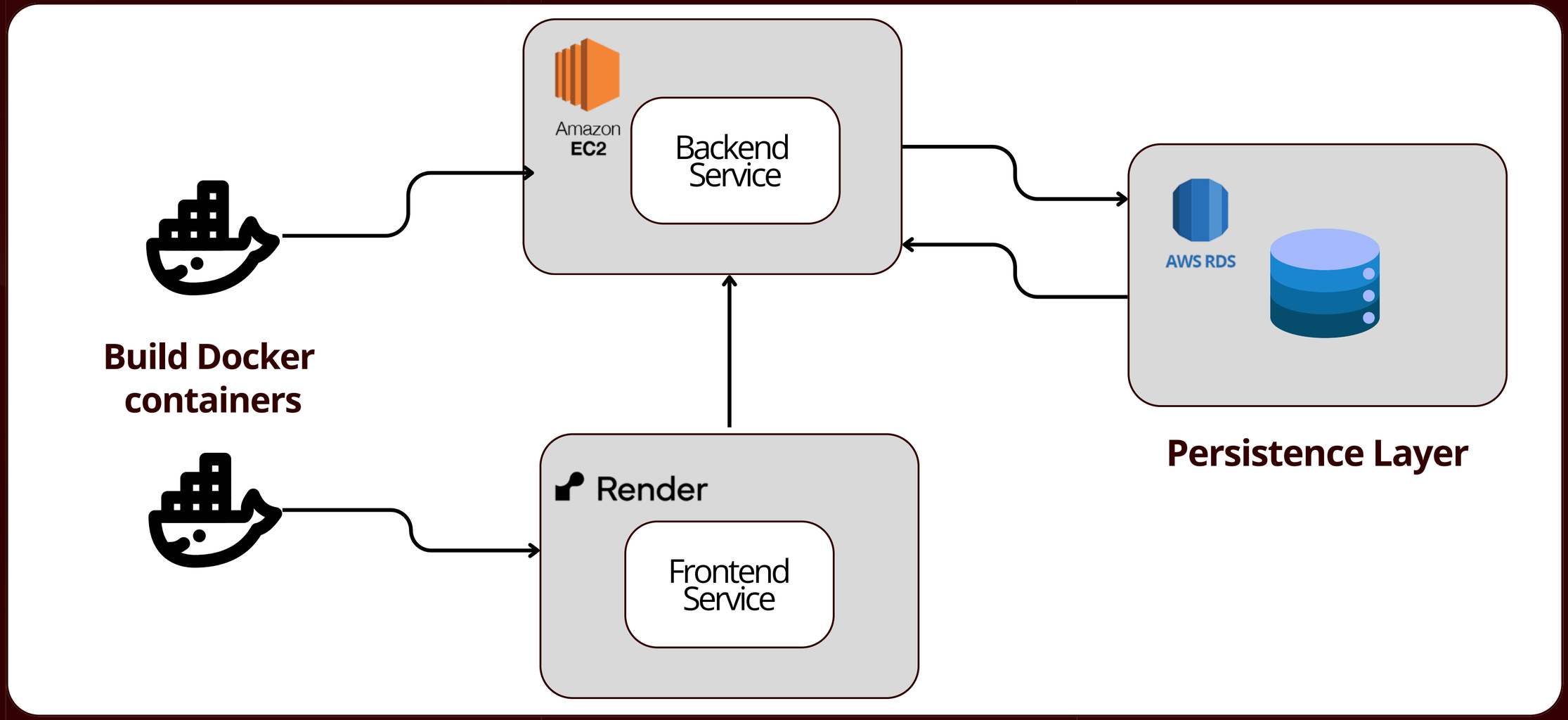
MariaDB



Docker

METHODOLOGY

Deployment



METHODOLOGY

Core Features & Demo

Access the application by using this URL:

https://indoor-navigation-ui.onrender.com/

Or scan this QR code:

METHODOLOGY

Core Features & Demo

1. **Indoor Navigation**
- Allow the user to select the room origin and destination and generate a route

2. **Layout Builder**
- Add, edit, and arrange rooms, hallways, and other components to build a floor layout
- Includes a draft and publish map workflow

RESULTS AND DISCUSSION

Navigation Accuracy and Application Usability

TABLE I: Raw Scores per Respondent

Respondent	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10
R1	4	1	5	1	5	3	5	5	5	5
R2	3	1	5	1	4	3	4	1	5	1
R3	4	2	4	2	5	3	5	4	4	4
R4	5	1	5	1	5	1	5	1	5	2
R5	5	1	5	1	5	1	5	1	5	1
R6	3	1	4	3	4	1	4	2	4	5

Respondent	SUS Score	Rating
R1	72.5	Good
R2	85.0	Excellent
R3	67.5	OK
R4	97.5	Excellent
R5	100	Excellent
R6	67.5	OK

Mean SUS Score: 81.67, which falls under the"Excellent" rating

All six respondents reached their intended destination, and routes were rated accurate to mostly accurate. Two respondents (R3, R6) scored lower with both scores at 67.5, likely reflecting the routing issues reported in their feedback

Strongest-rated aspects: ease of use, functional integration, and learnability (each ≥4.6/5 average across respondents)

RESULTS AND DISCUSSION

Layout Builder Usability

- Administrators found the layout builder intuitive for early-stage planning.
- One respondent compared it to a bubble diagram for visualizing room connections before committing to a final layout.
- Overall sentiment says that using the layout builder could help communicate proposed spatial arrangements of building layouts to clients and stakeholders during planning discussions.

RESULTS AND DISCUSSION

Layout Builder Usability

Some issues identified:

1. No autosave feature: Accidental misclicks could cause unsaved progress to be lost entirely
2. Difficulty distinguishing the "Layout Builder" from the separate "Sandbox" feature
3. No real unit of measurement is available when adjusting the size of components
4. No option to directly upload a floor plan and convert to a map layout

Conclusion

- The two testing groups suggested that the application performed strongly on its core functions: indoor navigation and enabling administrators to build working map layouts. These findings support the application as a viable and scalable approach to indoor navigation.

- Limitations include the small sample size (six end users, four administrators) and single-session testing, which limits insight into repeated or long-term use.

Future Work

01. Multi-Building Expansion — extend coverage across the campus, beyond a single building

02. Native Mobile Apps — iOS and Android versions for a better on-the-go experience

03. Accessibility Routing — PWD-accessible paths (ramps, elevators, accessible restrooms) and emergency exit routes

04. Real-Time Location Tracking — continuous position updates during navigation, rather than a static route

05. Layout Builder Refinements — autosave, real-world unit scaling, and floor plan file uploads

06. Crowdsourced Feedback — let users report map inaccuracies directly within the app

Thank You

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